

UDAV: Uncertainty-Driven Adaptive VLM Waypoint Planner

Ghazal Farhani* and Shabnam Shabani

Abstract—A key challenge in using vision–language models (VLMs) for aerial off-road navigation is that their predicted routes provide no indication of reliability. We propose UDAV, an *Uncertainty-Driven Adaptive VLM Waypoint Planner* for UAV-guided UGV navigation. UDAV generates repeated stochastic trajectories from the VLM, selects their medoid as the nominal self-consistent route, and uses their spatial dispersion to estimate predictive uncertainty. If the maximum uncertainty among interior waypoints exceeds a threshold, the planner invokes an additional reconsideration stage; otherwise, it returns the medoid directly.

Across 400 held-out trajectory queries from two UAV flights, stochastic medoid selection reduces mean ADE from 147.4 px for a deterministic VLM prediction to 115.9 px. The complete UDAV planner achieves a mean ADE of 110.4 px, a 25.1% reduction, with valid trajectories for all 400 queries. UDAV also achieves the lowest P90 and P95 errors among all evaluated configurations, including a higher-budget $K = 10$ consensus baseline. Relative to the $K = 5$ medoid, it reduces P90/P95 ADE from 225.3/326.0 px to 199.0/290.8 px. Thus, stochastic VLM predictions provide both a stronger nominal route and an actionable uncertainty signal for selectively mitigating large planning errors.

Index Terms—Vision Language Model, UAV-UGV collaboration.

I. INTRODUCTION

Autonomous navigation in unstructured off-road environments, with applications in search and rescue, disaster response, and remote inspection [1], has remained a fundamental challenge in field robotics. Unlike structured roads or indoor environments, off-road scenes lack explicit traversability boundaries, and the navigability of visually ambiguous terrain often cannot be inferred from geometry alone. Unmanned aerial vehicle (UAV)–unmanned ground vehicle (UGV) collaboration can address these challenges. The UGV keeps local sensing and control, while a UAV can quickly acquire a global view of the environment, spot hazards beyond the UGV’s field of view, and provide high-level route guidance [2], [3].

In a classical aerial-planning pipeline, first semantic or traversability information is inferred, converted into a cost map, and then a route with a classical planner such as A^* [4], [5] is proposed. This works well when terrain classes and traversal costs are known in advance, but it requires a pre-defined semantic class set, an explicit class-to-cost mapping, and environment-specific supervision, restrictive assumptions in previously unseen or highly variable off-road terrain.

Vision–language models (VLMs) offer an alternative as they associate visual observations with semantic and spatial concepts and can directly produce structured navigation outputs [6], [7]. Recent work has begun applying them to embodied navigation [8], [9]. However, aerially guided off-road planning with locally deployable VLMs, remains largely unexplored, especially when the planner must judge the reliability of a predicted trajectory before execution.

A single deterministic VLM prediction says nothing about its own reliability. Under stochastic decoding, the same scene can produce substantially different waypoint sequences, but a deterministic prediction hides this variability and offers no basis for deciding whether additional reasoning is warranted. In this study, we ask whether disagreement among stochastic trajectory predictions can serve as an actionable signal for when an otherwise self-consistent plan should be reconsidered.

To answer this, we develop UDAV, an uncertainty-driven adaptive VLM waypoint planner for UAV-guided off-road navigation. A locally deployable VLM is fine-tuned to map an aerial image, a start location, and a target location to an ordered sequence of N image-space waypoints. At inference time the planner draws K stochastic trajectory predictions and uses them twice: their medoid becomes the nominal self-consistent trajectory, and the per-waypoint covariance across the samples yields a geometric estimate of predictive uncertainty. Consensus and uncertainty thus come from the same samples, with no separate uncertainty-estimation model. If the maximum normalized uncertainty stays below a predefined threshold, UDAV returns the stochastic medoid directly; if it is exceeded, the model is told that its previous predictions disagreed within the identified region and is prompted to reconsider the route, and the final trajectory is taken as the medoid of the reconsidered predictions.

Two empirical observations motivate this design. Stochastic trajectory disagreement is significantly associated with trajectory error, and the benefit of stochastic sampling extends beyond uncertainty estimation: medoid-based self-consistency alone substantially improves accuracy over a single deterministic prediction. In UDAV, uncertainty consequently plays a distinct role, not defining the nominal trajectory, but deciding when an already self-consistent plan warrants a second planning pass.

We evaluate UDAV on two flight sequences from the GA3T UAV–UGV off-road dataset [10]: 193 spatially separated aerial observations with five candidate UGV paths each, giving 965 annotated path examples, of which a fixed held-out test set of 400 trajectory queries is used for evaluation. Deterministic

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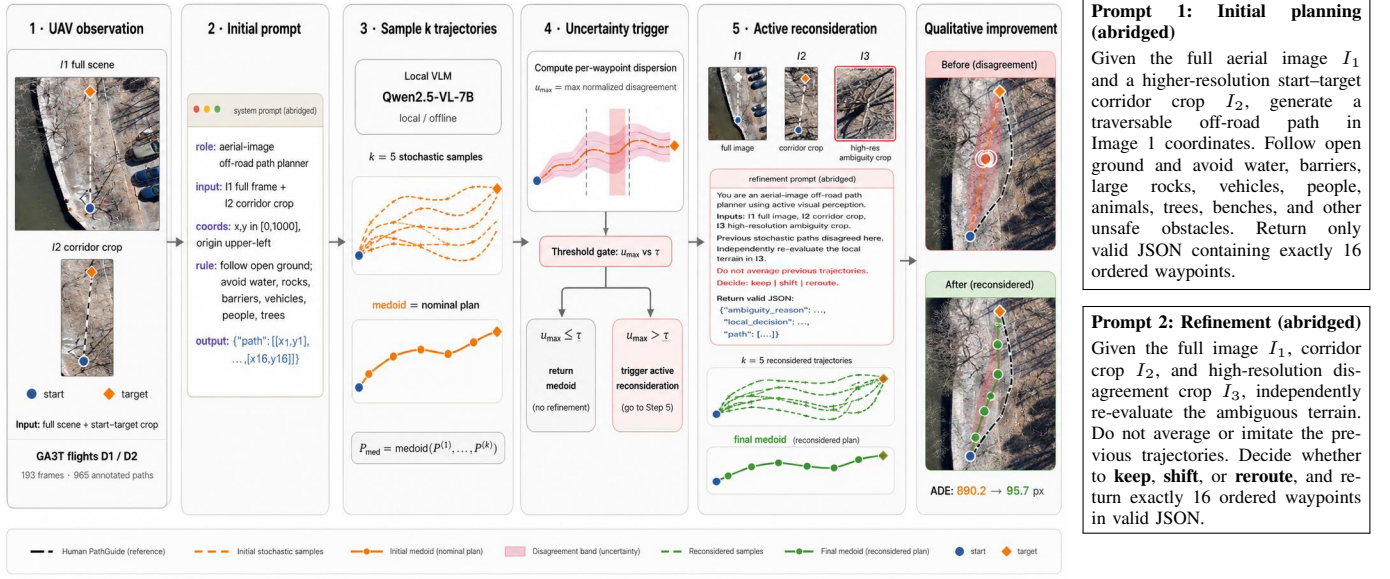


Fig. 1. Schematic of the proposed UDAV planner, with the initial planning and uncertainty-triggered refinement prompts shown in abridged form. The complete prompts used in the experiments are included in the released reproducibility package.

VLM planning reaches a mean average displacement error (ADE) of 147.4 px on this test set; selecting the $K = 5$ stochastic medoid brings this down to 115.9 px, and the full UDAV planner at $\tau = 0.6$ to 110.4 px, a 25.1% reduction relative to deterministic planning, with valid trajectories for all 400 queries. On the 174 queries where reconsideration is activated, mean ADE drops from 153.7 px to 141.0 px, and over the full test set UDAV attains the lowest P90 and P95 errors of all evaluated configurations, below even a higher-budget $K = 10$ consensus baseline.

This work makes three contributions (the overall planner is shown in Fig. 1). First, we formulate stochastic VLM trajectory disagreement as a geometric predictive-uncertainty signal for aerially guided off-road waypoint planning. Second, we introduce an adaptive planning mechanism that uses medoid self-consistency as the nominal prediction and invokes a second VLM planning pass only when trajectory disagreement exceeds a prescribed threshold. Third, we show that stochastic sampling provides two complementary benefits: medoid self-consistency improves nominal trajectory accuracy, while uncertainty-triggered reconsideration reduces large-error failures. To facilitate reproducibility, we release the ground-truth annotations used for fine-tuning and evaluation, the final fine-tuned model weights, and detailed data-preparation and reproduction instructions in the accompanying reproducibility package.

II. RELATED WORK

A. Vision-Language Models for Navigation

Because VLMs emit text while control requires continuous coordinates, a recurring strategy is to render candidate actions in the image and let the model choose among them. *Nasiriany et al.* [11] formalise this as PIVOT, which casts spatial

reasoning as iterative visual question answering: candidates are drawn on the image, the VLM selects the most promising, and a refined distribution is resampled and requiered until the answer converges. *Sathyamoorthy et al.* [12] apply the same principle to navigation in CoNVOI, overlaying numbered markers on obstacle-free regions of a ground-level RGB image and prompting cloud-hosted VLMs (GPT-4V, Gemini) to select the markers satisfying a context-dependent behaviour phrase. Their ablation is instructive: removing the markers raises the proportion of unacceptable paths from 0% to 85.7%, indicating that current VLMs cannot reliably reason about free space without an explicit spatial prior. Both methods consequently narrow an externally supplied proposal set rather than generating a route directly, and the authors of CoNVOI identify the absence of a global top-down view as a limitation to be addressed in future work.

A different composition is used by *Shah et al.* [13] in LM-Nav, which combines three pre-trained models without fine-tuning: an LLM parses the instruction into landmark names, CLIP grounds those names in observations, and a goal-conditioned policy (ViNG) executes the motion between them. Language thus determines *which* landmarks to visit, while the route between them is produced by a separate visual policy. LM-Nav also depends on a pre-collected traversal graph, and therefore does not operate in previously unobserved terrain.

B. Traversability Estimation for Off-Road Navigation

Classical off-road pipelines infer terrain properties and convert them into a planning cost. *Guan et al.* [14] proposed GA-Nav, which segments unstructured outdoor scenes into coarse traversability groups and assigns each group a traversal cost, producing smoother routes than pixel-wise semantic segmentation. *Shaban et al.* [15] similarly classify terrain into

predefined semantic classes for off-road driving. Both require the class set and the class-to-cost mapping to be specified in advance, and both are specific to the platform and the environment. An alternative line learns traversability from experience rather than from labels: WayFAST [16] supervises a traversability predictor using traction estimates recorded during driving, which removes the need for hand-defined costs but presumes prior traversal of representative terrain.

C. UAV-UGV Collaboration

Aerial-ground teaming has been studied extensively, spanning inter-vehicle communication, task allocation, and navigation optimisation; [17] provides a comprehensive review. Within this body of work, however, the use of VLMs for aerially guided ground navigation remains largely unexplored. Most closely related to our setting, *Cladera et al.* [7] demonstrate that decomposing an air-ground mission into interdependent sub-tasks, relayed between the UAV and the UGV, supports effective autonomous operation. Their emphasis is on coordinating and sequencing these sub-tasks; by contrast, we task the language model with producing the waypoint sequence that carries the UGV to an assigned target, together with an explanation of the visual evidence supporting it.

III. METHOD

UDAV uses stochastic VLM predictions: their medoid provides a self-consistent nominal trajectory, while their geometric dispersion determines whether additional visual reasoning is required. The overall procedure is illustrated in Fig. 1.

A trajectory is represented by $N = 16$ ordered image-space waypoints, $W = (w_1, \dots, w_N)$, where $w_i = [x_i, y_i]^\top$. Given the full aerial image I_1 , corridor crop I_2 , start s , and target t , the fine-tuned VLM generates K stochastic trajectories,

$$W^{(k)} \sim \pi_\theta(W \mid I_1, I_2, s, t), \quad k = 1, \dots, K, \quad (1)$$

with $K = 5$ in our experiments.

A. Stochastic Consensus

Trajectory similarity is measured as

$$D_T(W^{(a)}, W^{(b)}) = \frac{1}{N} \sum_{i=1}^N \left\| w_i^{(a)} - w_i^{(b)} \right\|_2. \quad (2)$$

The nominal trajectory is selected as the medoid,

$$W^{\text{med}} = W^{(k^*)}, \quad k^* = \arg \min_k \frac{1}{K-1} \sum_{\substack{j=1 \\ j \neq k}}^K D_T(W^{(k)}, W^{(j)}). \quad (3)$$

Unlike pointwise averaging, the medoid remains one of the trajectories generated by the VLM.

B. Uncertainty-Triggered Reconsideration

For each interior waypoint, the empirical covariance across stochastic predictions is

$$\Sigma_i = \frac{1}{K-1} \sum_{k=1}^K \left(w_i^{(k)} - \mu_i \right) \left(w_i^{(k)} - \mu_i \right)^\top, \quad (4)$$

where μ_i is the corresponding sample mean. The normalized uncertainty is then defined as

$$u_i = \frac{\sqrt{\chi_{2,0.95}^2 \lambda_{\max}(\Sigma_i)}}{\|t - s\|_2}, \quad i = 2, \dots, N-1. \quad (5)$$

If $\max_i u_i \leq \tau$, the nominal medoid is returned directly. Otherwise, uncertain waypoints are grouped into maximal contiguous regions, and the region containing the largest uncertainty is selected. Optionally, a high-resolution crop I_3 of this region is supplied to the VLM together with I_1 and I_2 ; the core method operates without this additional crop, and its effect is evaluated separately in Sec. IV-D. The refinement prompt informs the model that its previous predictions disagreed and asks it to independently reassess the local terrain rather than averaging the previous trajectories (the right panel of Fig. 1 shows the prompt).

The model generates $K_{\text{ref}} = 5$ reconsidered trajectories,

$$\widetilde{W}^{(q)} \sim \pi_\theta(W \mid I_1, I_2, I_3, s, t, \mathcal{C}_\tau), \quad q = 1, \dots, K_{\text{ref}}, \quad (6)$$

where $\mathcal{C}_\tau$ denotes the uncertainty cue and I_3 is included only in the crop-augmented variant. Their medoid, W^{ref} , is selected using the same trajectory-distance criterion in Eq. (2). The final planner is therefore

$$W^{\text{final}} = \begin{cases} W^{\text{med}}, & \max_i u_i \leq \tau, \\ W^{\text{ref}}, & \max_i u_i > \tau. \end{cases} \quad (7)$$

C. Dataset Overview

We used two datasets from the GA3T repository: 20260318-140904-dataset, referred to as **D1**, and 20260318-141807-dataset, referred to as **D2** [10]. Because both the UAV and UGV move relatively slowly, consecutive UAV frames are often highly similar and provide limited additional information for VLM-based path learning. We therefore subsampled each sequence to retain frames that were visually distinct while still preserving the progression of the UGV along the trail.

D1 originally contained 790 UAV frames, from which 99 representative frames were selected. For each selected frame, five human-annotated **PathGuide trajectories** were generated. One trajectory represented the primary forward direction of travel along the trail and was used to preserve correspondence with the route subsequently followed by the UGV. The remaining four trajectories represented alternative feasible paths through the same scene, increasing the diversity of the fine-tuning data and exposing the model to more challenging navigation alternatives. This resulted in a total of 495 trajectory samples. D1 was partitioned at the frame level

into 69 training frames (345 trajectories) and 30 permanently held-out test frames (150 trajectories). To reduce information leakage arising from temporally adjacent and visually similar UAV observations, consecutive frames were first grouped into blocks of 10, and the train-test partition was performed at the block level. Consequently, all five PathGuide trajectories associated with a given frame remained within the same partition.

The same frame-selection strategy was applied to D2. From the original 850 UAV frames, 94 representative frames were retained, producing 470 PathGuide trajectory samples. A fixed set of 50 complete frames, corresponding to 250 trajectories, was reserved as the permanent D2 test set, while the remaining 44 frames (220 trajectories) were used exclusively for domain adaptation.

In addition to the PathGuide trajectories, scene objects and relevant terrain features in each selected frame (including rocks, water, traversable path regions, vehicles, and other obstacles) were manually annotated using **LabelMe**. These annotations served two purposes: they provided supervision for fine-tuning the semantic segmentation model, and they were used to construct the obstacle-aware cost maps required by the A* planner, which is used as a classical method for comparison against our results. The resulting trajectory annotations are available in the PathGuide annotation dataset.

1) *VLM model*: Initially, without any fine-tuning, we evaluated four VLMs on a subset of the test data: Qwen2.5-VL-7B, InternVL3-8B, LLaVA-OneVision-7B, and SmolVLM2-2.2B. Among these models, Qwen2.5-VL-7B and InternVL3-8B achieved the lowest waypoint prediction error compared with the ground-truth trajectories. In preliminary fine-tuning experiments, Qwen2.5-VL-7B adapted substantially more effectively than InternVL3-8B; all subsequent experiments therefore use Qwen2.5-VL-7B. A further advantage of Qwen2.5-VL-7B is that it can be deployed offline, making it particularly useful in environments with limited or no internet connectivity. The prompt is a simple text instruction asking the model to use the visual information to determine the best path from the UGV to the target and generate 16 waypoints along that path.

D. A* Planning

A* [5] is a graph-search algorithm that recovers a minimum-cost path between a start and a goal node while excluding regions deemed non-traversable, and it remains widely used in ground- and aerial-robot planning. In grid-based implementations the environment is encoded as a cost map in which each cell carries a traversal cost derived from obstacle occupancy, terrain type, and a desired clearance margin around hazardous regions. The recovered path therefore reflects the cost map as much as the search itself.

We fine-tune `segformer-b5` [18] on the 113 non-test training/adaptation frames (69 from D1 and 44 from D2) over nine classes: background, water, barrier, vehicle, person, animal, rock, bench, and tree. The resulting semantic predictions are converted into the obstacle-aware cost map used by the

A* planner. No permanent-test annotation is used to train the segmentation model.

E. Metrics of Comparison

We evaluate geometric similarity between the predicted and reference trajectories using four complementary image-space metrics:

- **ADE (Average Displacement Error)**: the mean Euclidean distance between corresponding points of the resampled predicted and reference trajectories.
- **MidADE**: ADE computed over the interior waypoints only, excluding the fixed start and target endpoints.
- **Chamfer Distance**: a symmetric nearest-neighbor distance between the two trajectory point sets.
- **Hausdorff Distance**: the maximum nearest-neighbor mismatch between the two trajectory point sets.

Lower values indicate greater geometric similarity.

F. Fine-Tuning Setup

The model is first adapted on the D1 training partition, which contains 70% of the selected D1 samples, using QLoRA with 4-bit NF4 quantization and double quantization. Training is performed for three epochs with paged 8-bit AdamW, a cosine learning-rate schedule, an initial learning rate of 2×10^{-4} , and an effective batch size of eight. Computation is carried out in BF16. As described in Sec. III-C, the training and test samples are spatially separated to reduce leakage between temporally and spatially adjacent UAV observations. The resulting D1-adapted model is evaluated on both the held-out D1 test set and D2, which corresponds to a different UAV flight and environment. The model is then further adapted on D2 for one epoch using a reduced learning rate of 5×10^{-5} and reevaluated on both test sets. All runs are performed in Google Colab on an NVIDIA A100 80 GB GPU, using BF16 computation and a maximum of 768 visual tokens per image.

IV. RESULTS

A. Sequential Fine-Tuning and Cross-Flight Generalization

Table I summarizes the trajectory prediction performance of Qwen2.5-VL-7B before and after sequential fine-tuning.

Fine-tuning on D1 substantially improves performance relative to the zero-shot model. On D1, ADE decreases from 241.8 px to 149.3 px, while Chamfer distance decreases from 140.0 px to 75.8 px. Importantly, the D1-adapted model also transfers to the unseen D2 flight, reducing ADE from 218.8 px to 194.7 px without exposure to D2 during training. This demonstrates that the learned waypoint representation exhibits cross-flight generalization, although a noticeable domain gap remains.

Subsequent adaptation on D2 further reduces D2 ADE from 194.7 px to 156.8 px, corresponding to a 19.5% improvement relative to the D1-only model. Adaptation to D2 does not degrade performance on D1; instead, D1 ADE further decreases from 149.3 px to 131.7 px, with corresponding improvements in Mid-ADE, Chamfer distance, and Hausdorff distance. Consequently, sequential D1→D2 adaptation reduces

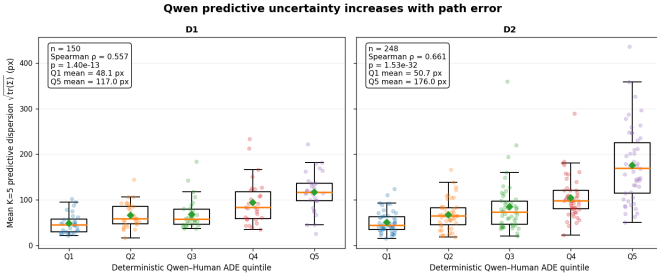


Fig. 2. **Predictive disagreement is associated with trajectory error.** Test cases are binned by trajectory-error quintile and plotted against $K = 5$ predictive dispersion. Dispersion increases across error groups in both flights (Spearman $\rho = 0.557$ on D1 and 0.661 on D2).

the combined ADE from 177.8 px to 147.4 px, with no evidence of catastrophic forgetting in the evaluated setting.

B. Stochastic Consensus and Uncertainty as a Reliability Signal

Before evaluating the complete adaptive planner, we examine the two quantities on which UDAV is built: stochastic consensus and predictive disagreement.

Table II reports the computational cost of stochastic sampling. Because multiple trajectories are generated within the same inference call, latency grows sublinearly with the number of samples. Increasing from $K = 1$ to $K = 5$ raises mean latency from 12.95 s to 15.57 s, an increase of approximately 20%, while $K = 10$ requires 18.51 s. We therefore use $K = 5$ as the default stochastic sampling budget for the uncertainty-driven planner and separately evaluate $K = 10$ as a higher-compute consensus baseline.

More importantly, stochastic disagreement is informative about trajectory reliability. Using $K = 5$, stochastic trajectories are generated for the complete permanent test sets. Figure 2 groups cases by trajectory error and shows that predictive dispersion increases with error. At the case level, predictive uncertainty is positively correlated with trajectory error, with Spearman correlations of $\rho = 0.557$ on D1 and $\rho = 0.661$ on D2; both associations are statistically significant.

This result is central to UDAV: disagreement among repeated VLM predictions is not merely sampling noise, but provides an online signal of planning difficulty that can be used to decide when additional reasoning is warranted.

C. Consensus Planning and Uncertainty-Guided Reconsideration

Table III summarizes the principal planning results over all 400 permanent-test queries. A single deterministic VLM prediction achieves a mean ADE of 147.4 px and produces valid trajectories for 398 of the 400 queries.

Selecting the medoid of only $K = 5$ stochastic predictions substantially improves this result, reducing mean ADE from 147.4 px to 115.88 px while producing valid trajectories for every query. This 21.4% reduction demonstrates that stochastic self-consistency alone provides a substantially stronger nominal planner than a single deterministic VLM generation.

Increasing the stochastic sampling budget to $K = 10$ provides an additional improvement, reducing mean ADE to 108.58 px and median ADE to 80.37 px. This establishes that larger-sample consensus can further improve nominal trajectory accuracy. However, $K = 10$ requires the larger sampling budget for every query.

UDAV instead uses the $K = 5$ ensemble both to construct the nominal medoid and to determine when additional reasoning is required. The threshold sweep in Table III illustrates the resulting accuracy-computation trade-off. At $\tau = 0.6$, reconsideration is activated for 174 of 400 queries (43.5%), while the remaining 56.5% return the initial stochastic medoid directly.

At this operating point, UDAV achieves a mean ADE of 110.36 px, close to the 108.58 px obtained by the $K = 10$ medoid. With $K_{\text{ref}} = 5$, the selective policy corresponds to an average sampling budget of $5 + 0.435(5) = 7.18$ generated trajectories per query, compared with 10 trajectories for the $K = 10$ consensus baseline.

The distinction between the two strategies becomes particularly interesting in the high-error tail. Although the $K = 10$ medoid achieves the lowest mean ADE, UDAV at $\tau = 0.6$ achieves slightly lower P90 and P95 errors: 198.98 px and 290.78 px, compared with 204.92 px and 292.01 px for the $K = 10$ medoid. Thus, larger-sample consensus primarily improves average trajectory accuracy, whereas uncertainty-guided reconsideration provides a targeted mechanism for addressing difficult cases without applying the larger inference budget uniformly.

On the 174 queries that activate reconsideration at $\tau = 0.6$, the initial $K = 5$ medoid has a mean ADE of 153.71 px, compared with 141.01 px after reconsideration, corresponding to a mean reduction of 12.71 px. Reconsideration improves 50.0% of the triggered trajectories. The paired one-sided Wilcoxon test is not significant ($p = 0.335$), and the bootstrap confidence interval for the all-query mean improvement includes zero. The second planning stage should therefore not be interpreted as a guaranteed improvement for every uncertain query; rather, its benefit is concentrated in a subset of difficult cases.

The threshold sweep also shows that reconsidering more trajectories does not necessarily improve performance. At $\tau = 0.3$, 71.0% of queries are escalated, yet the resulting mean ADE is 115.03 px, only marginally better than the $K = 5$ medoid. Conversely, thresholds above approximately 1.0 trigger too few reconsiderations to provide a consistent benefit. The useful operating region therefore lies between these extremes, illustrating that the uncertainty threshold controls a meaningful trade-off between additional computation and trajectory accuracy. We note that τ was selected from this sweep rather than calibrated on a separate validation split; mean ADE varies by less than 4 px across $\tau \in [0.5, 0.8]$, although the tail percentiles are more sensitive to the threshold.

Qualitative examples in Fig. 4 illustrate the behavior motivating the adaptive stage. In these examples, stochastic predictions disagree strongly around locally ambiguous regions and the second planning pass substantially modifies the selected

TABLE I
TRAJECTORY PREDICTION PERFORMANCE OF QWEN2.5-VL-7B BEFORE AND AFTER SEQUENTIAL FINE-TUNING.

Dataset	Model Condition	ADE (px) ↓	Mid-ADE (px) ↓	Chamfer (px) ↓	Hausdorff (px) ↓	Valid Path (%) ↑
D1	Qwen (zero-shot)	241.8	250.3	140.0	435.7	94.7
D1	Qwen after D1 fine-tuning	149.3	169.7	75.8	202.9	99.3
D1	Qwen after D1→D2 fine-tuning	131.7	149.6	74.5	167.1	100.0
D2	Qwen (zero-shot)	218.8	225.8	130.4	411.7	92.4
D2	Qwen after D1 fine-tuning	194.7	219.8	101.9	253.0	100.0
D2	Qwen after D1→D2 fine-tuning	156.8	177.7	74.7	175.6	99.2
Combined	Qwen (zero-shot)	227.6	235.1	134.1	420.9	93.2
Combined	Qwen after D1 fine-tuning	177.8	201.1	92.1	234.3	99.8
Combined	Qwen after D1→D2 fine-tuning	147.4	167.1	74.7	172.4	99.5

Lower ADE, Mid-ADE, Chamfer distance, and Hausdorff distance indicate greater geometric similarity to the ground-truth trajectory. Valid Path denotes the percentage of predictions that could be parsed as valid trajectories.

TABLE II
INFERENCE LATENCY FOR DIFFERENT NUMBERS OF STOCHASTIC SAMPLES.

K	Mean	Median	Time/Path	Relative
1	12.95	13.65	12.95	1.00×
3	14.64	15.11	4.88	1.13×
5	15.57	15.91	3.11	1.20×
10	18.51	19.00	1.85	1.43×

Latency values are in seconds. Time/Path denotes total generation time divided by the number of sampled trajectories.

TABLE III
FULL-TEST-SET WAYPOINT-PLANNING PERFORMANCE OVER 400 TRAJECTORY QUERIES. LOWER TRAJECTORY ERRORS ARE BETTER.

Method	Valid	Mean ADE	Median	P90	P95	Reconsider.
Deterministic VLM	99.5%	147.4	92.6	359.5	—	—
$K = 5$ stochastic medoid	100.0%	115.88	84.06	225.31	326.04	—
$K = 10$ stochastic medoid	100.0%	108.58	80.37	204.92	292.01	—
UDAV, $\tau = 0.30$	100.0%	115.03	87.92	227.02	326.89	71.0%
UDAV, $\tau = 0.50$	100.0%	109.58	84.49	202.13	297.48	49.0%
UDAV, $\tau = 0.60$	100.0%	110.36	83.69	198.98	290.78	43.5%
UDAV, $\tau = 0.70$	100.0%	113.41	82.59	227.38	326.04	38.5%
UDAV, $\tau = 0.80$	100.0%	110.32	83.18	213.90	322.30	35.8%
UDAV, $\tau = 1.00$	100.0%	116.66	86.71	227.02	327.82	30.0%
UDAV, $\tau = 1.20$	100.0%	116.09	84.06	225.75	326.04	26.5%

ADE values are in pixels. “Reconsider.” denotes the fraction of queries for which uncertainty exceeds τ and triggers the second planning stage. The $K = 10$ medoid achieves the lowest mean ADE, while UDAV at $\tau = 0.60$ achieves the lowest P90 and P95 errors.

route. These examples complement the aggregate statistics by illustrating that the benefit of reconsideration is concentrated in difficult planning cases rather than uniformly distributed across the test set.

D. Visual Re-observation Ablation

We additionally investigate whether supplying extra visual detail during reconsideration improves the refined trajectory.

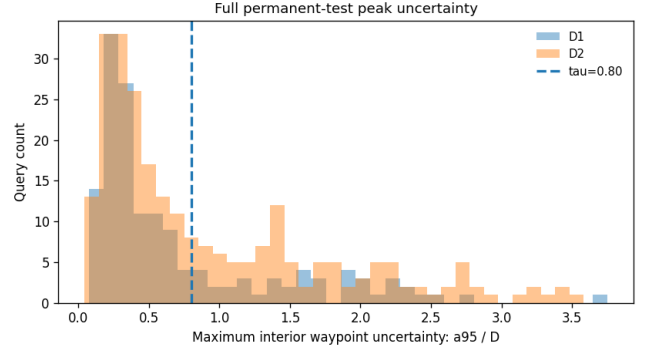


Fig. 3. Distribution of the maximum normalized interior-waypoint uncertainty $u_{\max}$ over the D1 and D2 permanent test sets. At $\tau = 0.6$, 174 of 400 queries (43.5%) trigger reconsideration.

This ablation was performed before the final operating point was fixed, on the 143 queries triggered at the stricter threshold $\tau = 0.8$; the comparison among the crop variants is internal to this subset and does not depend on the choice of τ elsewhere in the paper.

Reconsideration without an additional image achieves a mean ADE of 144.6 px. A broad crop covering the uncertain region yields 150.3 px, whereas a tightly localized peak-uncertainty crop reduces the mean ADE to 138.0 px. Although the peak-local crop produces the lowest mean error, neither crop variant is significantly different from reconsideration without an additional crop under the paired Wilcoxon tests.

These results suggest that localized visual re-observation can provide useful additional terrain detail, while broad re-observation does not necessarily improve the decision. We therefore treat crop design as a secondary component whose benefit is most apparent when additional visual information is concentrated around the region of greatest predictive disagree-

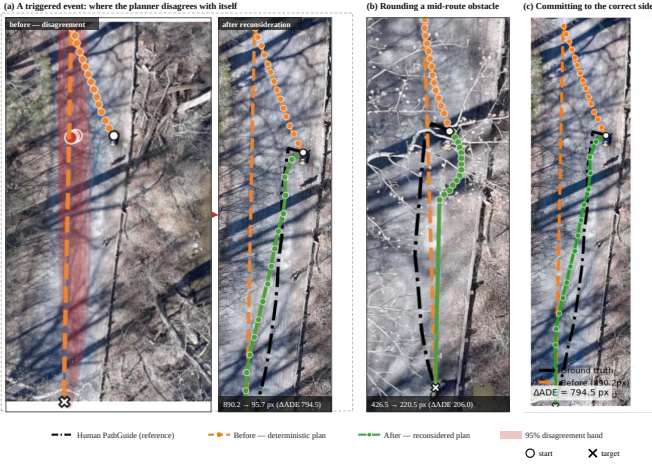


Fig. 4. **Qualitative examples of uncertainty-triggered reconsideration.** Stochastic trajectory disagreement identifies difficult regions in which an additional planning pass can modify the route toward the human-annotated reference.

ment.

E. Comparison with the Physically Executed UGV Trajectory

We finally compare the underlying VLM planner with the trajectory physically executed by the UGV. This experiment uses PathGuide 1, which corresponds to the actual forward route followed by the UGV during data collection. The remaining four PathGuides associated with each UAV frame define alternative feasible navigation tasks toward different target locations.

None of the queries that exceeded the uncertainty threshold originated from PathGuide 1. Consequently, the adaptive reconsideration branch is not evaluated by this physical-path experiment. The comparison therefore evaluates the underlying VLM planning representation rather than providing a closed-loop validation of the complete UDAV policy.

This distinction is consistent with the qualitative behavior observed in Fig. 4. Many high-uncertainty cases arise when the target requires a substantial departure from the nominal forward direction, such as a lateral maneuver, T-turn, or return toward a target located behind the vehicle. Such maneuvers are not represented by the physically executed forward trajectory in PathGuide 1.

Table IV compares the human annotation, the Qwen planner, and SegFormer+A* against the physically executed UGV trajectory. On the combined data, Qwen achieves an ADE of 131.2 px, compared with 124.2 px for the human PathGuide and 146.7 px for SegFormer+A*. On D2, Qwen obtains the lowest ADE and Chamfer distance among the three planning representations.

These results indicate that the VLM-generated trajectories remain geometrically consistent with physically feasible off-road motion, despite being produced directly from aerial imagery rather than through an explicitly constructed semantic cost map. A* remains substantially faster computationally; the purpose of this comparison is therefore not to demonstrate



Fig. 5. **Planned trajectories against the path physically executed by the UGV.** Two example scenes comparing the human PathGuide annotation, the Qwen planner, SegFormer+A*, and the executed UGV trajectory recovered by registering consecutive UAV frames. (a) D1. (b) D2.

computational superiority, but to assess how closely the different planning representations correspond to the trajectory that was physically executed by the UGV.

Representative examples are shown in Fig. 5.

V. CONCLUSION

We presented UDAV, an adaptive VLM waypoint planner that uses repeated stochastic trajectory predictions for both self-consistency and geometric uncertainty estimation. Selecting the medoid of $K = 5$ stochastic predictions already provides a substantially stronger nominal trajectory than a single deterministic VLM generation, reducing mean ADE from 147.4 px to 115.9 px over the complete permanent test set.

The same samples also yield a normalized geometric disagreement measure that escalates uncertain queries to a second planning pass. At $\tau = 0.6$, the complete policy achieves a mean ADE of 110.4 px, a 25.1% reduction relative to deterministic planning, with valid trajectories for all 400 evaluated queries. The benefit is concentrated in the high-error tail: UDAV attains the lowest P90 and P95 errors of all evaluated configurations, below even a $K = 10$ consensus baseline, at an average budget of 7.2 rather than 10 sampled trajectories per query. On the triggered queries themselves the paired improvement is not statistically significant, so the second pass acts as a selective failure-mitigation mechanism, not a guaranteed improvement for every uncertain query.

We also compared the underlying VLM trajectories with human annotations, a SegFormer+A* planner, and the trajectories physically executed by the UGV. Off-road scenes admit multiple feasible routes, and similarity to a single human annotation is not the only meaningful notion of planning quality. Because the reconsideration trigger did not activate on the physically executed PathGuide 1 routes, this experiment is contextual rather than a closed-loop evaluation of the adaptive policy.

TABLE IV
TRAJECTORY SIMILARITY WITH RESPECT TO THE PHYSICALLY EXECUTED UGV PATH.

Dataset	Path Compared with Actual UGV	ADE (px) ↓	Chamfer (px) ↓	Hausdorff (px) ↓
D1	Human	134.1 ± 68.5	34.4 ± 14.5	126.7 ± 5.0
D1	Qwen	145.6 ± 77.1	41.0 ± 16.8	143.8 ± 28.6
D1	SegFormer + A*	151.4 ± 72.1	42.7 ± 9.6	127.4 ± 5.7
D2	Human	104.4 ± 51.0	57.9 ± 16.3	137.0 ± 58.4
D2	Qwen	102.2 ± 59.3	48.8 ± 20.2	137.5 ± 68.1
D2	SegFormer + A*	137.2 ± 59.5	96.2 ± 43.9	178.5 ± 69.1
Combined	Human	124.2 ± 63.9	42.3 ± 18.6	130.1 ± 33.2
Combined	Qwen	131.2 ± 73.6	43.6 ± 18.0	141.7 ± 44.6
Combined	SegFormer + A*	146.7 ± 67.5	60.5 ± 36.3	144.4 ± 45.9

Lower values indicate greater similarity to the physically executed UGV trajectory.

The evaluation has two main limitations: only two flights from a single dataset and environment type were covered, and the reconsidered trajectories were not executed in closed loop. Future work will evaluate our proposed planner across more diverse terrain, and deploy the complete policy in closed-loop UGV execution.

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